



DATA ENGINEER INTERN

Talent Assessment Program · Technical Assignment

Abstract

This project analyzes Airbnb data to understand factors that influence listing performance, pricing, occupancy, and customer reviews. Data from listings, reviews, calendar records, and neighborhood information was collected, cleaned, and combined into a single dataset for analysis. Exploratory data analysis, statistical methods, and machine learning techniques were used to identify key trends and patterns within the Airbnb market. The findings provide useful insights for hosts and platform operators, helping them make better decisions related to pricing, customer satisfaction, and business performance.

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Executive Summary

This project focused on analyzing Airbnb marketplace data and developing a reliable data engineering and analytics workflow. Data from listings, reviews, calendar records, and neighbourhood information was collected, cleaned, standardized, and integrated into a unified dataset for analysis. Exploratory Data Analysis (EDA) was conducted to identify patterns in pricing, occupancy, host behavior, and customer reviews. Statistical analysis and machine learning techniques were then applied to better understand the factors influencing listing performance and revenue generation.

The analysis revealed that location, room type, occupancy levels, and review scores are among the most important factors affecting Airbnb listing performance. Listings with higher ratings and stronger occupancy generally achieved greater estimated revenue. The project also demonstrated the importance of maintaining high-quality data through proper validation, cleaning, and enrichment processes.

Based on the findings, Airbnb hosts should focus on competitive pricing, improving guest satisfaction, and maintaining strong review ratings to maximize performance. The developed data pipeline and analytical framework provide a scalable foundation for future business intelligence, forecasting, and decision-making applications.

Objectives & Scope

The primary objective of this project was to build an end-to-end data engineering and analytics solution using Airbnb data. The project aimed to transform raw datasets into clean, structured, and analytics-ready data while generating meaningful business insights through exploration analysis, statistical methods, and machine learning techniques.

The scope of the project included the analysis of Airbnb listings, reviews, calendar data, and neighbourhood information. Key activities involved data ingestion, profiling, cleaning, standardization, enrichment, and integration of multiple datasets into a unified master dataset. Exploratory Data Analysis was performed to understand pricing patterns, host characteristics, occupancy trends, and customer review behavior. In addition, a dimensional data model was designed to support analytical queries and reporting.

The project focused on identifying the factors that influence listing performance, occupancy, and revenue while providing actionable recommendations for hosts, investors, and platform operators. The analysis was limited to the available Airbnb datasets and did not include external economic, tourism, or demographic data sources.

Chapter 01

Dataset Familiarization

1.1 Dataset Source

The dataset used in this project was obtained from the Inside Airbnb public data repository, an independent initiative that provides publicly available information on Airbnb listings across cities worldwide. The dataset is commonly used for market analysis, policy research, and business intelligence applications.

For this study, a single city dataset was selected and downloaded from the Inside Airbnb platform. The extracted dataset consisted of four files:

- listings.csv
- calendar.csv
- reviews.csv
- neighbourhoods.csv

These files collectively provide information on listings, hosts, guest reviews, pricing, availability, and geographical neighbourhood classifications.

1.2 Business Domain Context

The Airbnb marketplace consists of several key entities that represent different aspects of the platform ecosystem.

Listing

A listing represents a property or accommodation advertised on Airbnb. Each listing contains information related to property characteristics, location, room type, host details, pricing information, and customer review metrics.

Host

A host is an individual or organization responsible for managing one or more listings. Hosts determine pricing strategies, availability schedules, booking policies, and guest interactions.

Review

The review represents feedback submitted by guests after completing a stay. Reviews provide qualitative insights into customer experiences and act as an indicator of listing quality and customer satisfaction.

Calendar

The calendar dataset stores daily availability and pricing information for each listing. This information can be used to estimate occupancy trends, booking patterns, and potential revenue opportunities.

Neighbourhood

Neighbourhood data provides geographical segmentation of listings and enables location-based analysis of market conditions, pricing patterns, and listing density.

1.3 Dataset Overview

The downloaded city dataset contained four files.

Dataset	Description
listings.csv	Core listing and host information.
reviews.csv	Guests review records and review text.
calendar.csv	Daily availability and pricing information.
neighbourhoods.csv	Geographic neighborhood reference data.

Listings Dataset

The listings dataset contains the primary information related to Airbnb properties and hosts. The dataset includes listing identifiers, descriptive attributes, geographical coordinates, room types, property characteristics, host information, review metrics, and availability indicators.

A total of 453 records and 85 attributes were identified within the dataset.

Reviews Dataset

The reviews dataset contains guest review information linked to individual listings. Each review includes reviewer information, review dates, and review comments.

The dataset contains 27,892 review records and 6 attributes.

Calendar Dataset

The calendar dataset stores daily listing availability and pricing information. Each record represents the availability status of a listing on a specific date.

The dataset contains 165,345 records and 7 attributes.

Neighbourhood Dataset

The neighbourhood dataset provides geographic reference information used for location-based analysis and grouping of listings.

The dataset serves as a supporting lookup table for geographical segmentation.

Detailed schemas, data types, ranges, and sample values for all datasets are provided in Appendix A.

1.4 Dataset Relationships

The four datasets are interconnected through listing identifiers.

The listings dataset serves as the central entity. Review records and calendar records are linked to listings through the listing identifier. The neighbourhood dataset provides location reference information associated with listings.

Primary Keys

Dataset	Primary key
listings	id
reviews	id
calendar	(listing_id, date)
neighbourhoods	Reference attributes

Foreign Keys

Dataset	Foreign key	References
reviews	listing_id	listings.id
calendar	listing_id	listings.id

listings	neighbourhood attributes	neighbourhoods
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1.5 Data Dictionary Review

The official Inside Airbnb data dictionary was reviewed to understand the meaning and intended use of key variables. Several attributes require special interpretation prior to analysis.

Price Variables

Price-related fields are stored as formatted strings containing currency symbols and separators. These values require standardization and conversion to numerical data types before analysis.

Availability Indicators

Variables such as `availability_365` and `available` indicate listing availability rather than confirmed bookings. An unavailable date may represent a guest reservation, host-imposed restriction, maintenance period, or other operational constraint.

Review Scores

Review score variables represent aggregated customer ratings and should be interpreted as subjective measures of guest satisfaction rather than objective indicators of property quality.

Superhost Status

The `host_is_superhost` attribute represents Airbnb's host quality designation and can be used as an indicator of host experience and performance.

1.6 Dataset Limitations

Several limitations were identified during the dataset exploration process.

Coverage Gaps

Several host-related attributes contain substantial missing values. Examples include host response metrics, acceptance rates, licensing information, and other operational indicators. These gaps limit the depth of host performance analysis.

Missing Historical Data

The dataset represents a snapshot of Airbnb activity rather than a complete historical record. Consequently, long-term market trends cannot be fully reconstructed from the available data alone.

Scraping Artifacts

As the dataset is generated through web scraping techniques, inconsistencies, missing values, outdated information, and formatting issues may be present.

Limited Demand Visibility

The dataset does not contain actual booking or revenue transactions. Demand and occupancy therefore must be inferred indirectly using review activity and availability patterns.

1.7 Assumptions

Several assumptions were established prior to analysis.

ID	Assumption
A1	Each listing ID uniquely identifies a property.
A2	Each review belongs to a single listing.
A3	Calendar records accurately represent listing availability at the time of data collection.
A4	Review counts can be used as a proxy measure of demand.
A5	Superhost status reflects higher host performance and reliability.

A6 Missing host metrics indicate unavailable information rather than zero values.

A7 Price values represent nightly accommodation rates.

These assumptions were documented to ensure consistency throughout the subsequent engineering, analytical, and modeling phases of the project.

1.8 Conclusion

The dataset provides a comprehensive representation of Airbnb marketplace activity through listings, reviews, availability records, and geographical information. Despite several limitations associated with missing values, web-scraped data quality issues, and the absence of direct booking information, the dataset offers sufficient coverage to support data engineering, exploratory analysis, statistical investigation, and predictive modeling tasks. The findings from this familiarization stage informed the design of the data engineering pipeline and subsequent analytical activities presented in later sections of this report.

Chapter 02

Engineering Approach

2.1 Overview

A robust data engineering pipeline was developed to transform raw Airbnb datasets into clean, validated, and analytics-ready datasets suitable for exploratory analysis, statistical testing, and machine learning applications. The engineering process focused on data quality, repeatability, scalability, and maintainability while ensuring that all transformations could be reproduced with minimal manual intervention.

The engineering workflow consisted of five major stages:

1. Data Ingestion and Profiling
2. Data Cleaning and Standardization
3. Data Enrichment and Integration
4. Data Modeling
5. Pipeline Automation and Metadata Management

The overall architecture is illustrated below:

Raw CSV Files

- Data Validation
- Data Cleaning
- Data Enrichment
- Analytical Data Model
- Machine Learning & Business Analytics

2.2 Data Ingestion and Profiling

The first stage involved ingesting the four extracted Airbnb datasets:

listings.csv

reviews.csv

calendar.csv

neighbourhoods.csv

The ingestion process was designed to be repeatable and capable of processing future city datasets with minimal modifications.

Dataset Profiling

Each dataset was profiled to understand its structure and quality characteristics. Profiling activities included:

- Row and column count analysis
- Data type inspection
- Column cardinality analysis
- Missing value assessment
- Duplicate detection
- Outlier identification
- Domain validation checks

The profiling process revealed that the listings dataset contained the highest number of attributes, while the calendar dataset contained the largest number of records due to its daily availability structure.

Data Quality Findings

Several data quality issues were identified:

- Significant missing values in host-related attributes
- Inconsistent formatting of price fields
- Mixed data types across selected attributes
- Missing geographical metadata in some records
- Potential duplicate listings requiring further investigation

A comprehensive data quality report was generated to document these findings and guide subsequent cleaning activities.

Duplicate Detection

Duplicate analysis was conducted using both deterministic and fuzzy matching techniques.

Deterministic duplicate detection was performed using exact record comparisons, while fuzzy matching was applied to textual listing names to identify potentially duplicated properties with minor naming variations.

The analysis identified a small number of potential duplicate candidates requiring manual verification.

Outlier Assessment

Outlier detection was performed using the Interquartile Range (IQR) method on key numerical variables including:

- Number of reviews
- Availability of indicators
- Accommodation capacity
- Bedroom count

Extreme values were reviewed individually to determine whether they represented genuine business observations or potential data quality issues.

Domain Validation

Several validation rules were applied:

- Latitude values must fall between -90 and +90.
- Longitude values must fall between -180 and +180.
- Review counts cannot be negative.
- Availability measures cannot exceed valid annual limits.
- Price values must be non-negative.

- Records violating these constraints were flagged for further inspection.

2.3 Data Cleaning and Standardization

Following profiling, a structured cleaning process was implemented to improve dataset consistency and reliability.

Price Standardization

Price-related attributes were stored as text strings containing currency symbols and separators. These fields were standardized by:

- Removing currency symbols
- Removing thousands of separators
- Converting values to numeric formats

This enabled numerical analysis and revenue calculations.

Date Standardization

Date fields within calendar and review datasets were converted into standardized datetime formats to support temporal analysis and feature engineering.

Categorical Normalization

Textual categorical fields such as room types and property types were standardized by:

- Removing whitespace inconsistencies
- Standardizing capitalization
- Resolving equivalent category names

This reduced category fragmentation and improved analytical consistency.

Missing Value Handling

Different missing value strategies were applied depending on the business meaning of the variable:

- Median imputation for selected numerical variables
- Explicit null preservation where missingness contained information
- Sentinel values for selected operational fields

Variables with excessive missingness were retained but documented as limitations.

Record Validation

Records failing validation rules were flagged and retained separately to maintain auditability and reproducibility.

Geographic Standardization

Neighbourhood names and coordinate fields were standardized to ensure consistent geographical analysis and aggregation.

2.4 Data Enrichment and Integration

To improve analytical value, multiple datasets were integrated into enriched analytical tables.

Listing Master Table

The listings dataset served as the primary entity and was enriched using aggregated review and calendar information.

The resulting master dataset contained:

- Listing attributes
- Host information
- Review statistics
- Availability indicators

- Geographic attributes
- Review Aggregation

Review records were aggregated at listing level to derive:

- Total review count
- Review activity indicators
- Review frequency metrics

These measures provide indirect indicators of demand and customer engagement.

Occupancy Estimation

Calendar availability data was transformed into estimated occupancy measures.

Occupancy rate was calculated as:

Occupancy Rate = Unavailable Days / Total Calendar Days

This provides an approximate indication of listing utilization.

Revenue Estimation

Estimated revenue was derived using availability and pricing information.

Estimated Revenue = Occupied Days × Nightly Price

As actual booking information was unavailable, these estimates should be interpreted as indicative rather than exact revenue figures.

Neighbourhood-Level Aggregates

Neighbourhood statistics were calculated to support market-level analysis, including:

- Average ratings
- Median pricing
- Listing density
- Review activity levels

These aggregated metrics enable comparison between different geographical areas.

Derived Features

Several new variables were engineered to improve analytical capability:

- Host tenure
- Review frequency
- Price per bedroom
- Estimated occupancy rate
- Revenue estimates
- Average review activity

These derived features were subsequently used in exploration analysis and predictive modeling.

2.5 Data Modeling

A dimensional data model was designed to support analytical workloads and future scalability.

Star Schema Design

The analytical warehouse was modeled using a star schema approach consisting of one central fact table and multiple dimension tables.

Fact Table

Fact_Listing_Metrics

The fact table stores measurable business metrics including:

- Occupancy estimates
- Revenue estimates
- Review counts
- Availability measures

Dimension Tables

Dim_Listing

- Listing attributes
- Property characteristics

Dim_Host

- Host information
- Superhost status

Dim_Neighbourhood

- Geographic attributes
- Location hierarchy

Dim_Date

- Calendar information
- Temporal attributes
- Key Relationships

Each dimension table is connected to the fact table using surrogate or business keys, enabling efficient analytical querying while reducing redundancy.

Modeling Considerations

The star schema was selected because:

- It simplifies analytical queries.
- It improves query performance.
- It supports future expansion.
- It aligns with common business intelligence practices.

A fully normalized model was considered but rejected due to increased query complexity and reduced analytical usability.

2.6 Pipeline Design and Automation

To ensure repeatability and maintainability, the engineering workflow was designed as a configurable pipeline.

Configurable Processing

Pipeline parameters such as file paths, city names, and output locations were externalized into configuration settings, allowing the same workflow to process future datasets with minimal modifications.

Logging and Monitoring

Logging mechanisms were implemented to track:

- File ingestion events
- Validation outcomes
- Processing errors
- Pipeline execution status

This improves traceability and operational reliability.

Error Handling

Exception handling was incorporated throughout the workflow to prevent pipeline failures caused by malformed files, missing attributes, or unexpected data formats.

Incremental Processing Strategy

An incremental processing approach was designed to support future dataset updates.

Rather than reprocessing all historical data, newly ingested records can be identified and processed independently using ingestion timestamps and unique identifiers.

This approach significantly reduces processing overhead and improves scalability.

Metadata Management

A metadata layer was designed to capture:

- Dataset source information
- Ingestion timestamps
- Processing timestamps
- Validation status
- Record counts

This metadata supports auditing, governance, and operational monitoring.

Data Lineage

Data lineage documentation was maintained to trace transformations from source files to analytical outputs.

Source Files

- Validation
- Cleaning
- Standardization

- Enrichment
- Master Dataset
- Dimensional Model
- Analytics & Machine Learning

This ensures transparency, reproducibility, and ease of maintenance.

2.7 Summary

The engineering pipeline successfully transformed raw Airbnb datasets into clean, validated, enriched, and analytics-ready data assets. Through systematic profiling, cleaning, enrichment, dimensional modeling, and automation, the resulting data infrastructure provides a reliable foundation for exploratory analysis, statistical investigation, machine learning experimentation, and business decision-making.

Summary of Completed Work

The project successfully implemented an end-to-end data engineering and analytics workflow using Airbnb datasets. Key deliverables included data ingestion, profiling, cleaning, standardization, enrichment, exploratory data analysis, statistical analysis, dimensional data modeling, and machine learning experimentation. A repeatable pipeline was developed to transform raw datasets into an analytics-ready master dataset, and business insights were generated through visualization and modeling techniques. The work was completed to a standard suitable for analytical reporting and decision-making purposes.

Summary of Incomplete Work

Due to time constraints and project prioritization, certain advanced components were not fully implemented. These included large-scale pipeline automation, advanced geospatial visualizations, integration of external datasets, and more sophisticated machine learning or AI-based forecasting models. Priority was given to establishing a reliable data foundation, completing the core engineering tasks, and generating meaningful business insights. This approach ensured that the most critical objectives of the assignment were successfully achieved while maintaining the quality and reliability of the completed work.

Key Lessons Learned

A key lesson from this project was the importance of data quality and preparation in the analytics lifecycle. Significant effort was required to understand, validate, and clean the data before meaningful analysis could be performed. The project also highlighted the value of combining technical analysis with business interpretation to produce actionable insights. Additionally, designing a structured workflow improved reproducibility and demonstrated the importance of planning and prioritization when working under limited time constraints.

Reflection

This project required balancing data engineering, analytics, and machine learning tasks within a limited timeframe. Work was prioritized by first building a reliable data foundation through data ingestion, profiling, cleaning, and validation before moving on to exploration and modeling. This approach ensured that subsequent analyses were based on accurate and consistent data.

Several trade-offs were made during the project. Due to time constraints, priority was given to developing a robust end-to-end pipeline and generating actionable business insights rather than implementing more advanced modeling techniques or integrating external data sources. Similarly, the dimensional data model was designed to be simple and practical for analytical use cases while remaining scalable for future enhancements.

One of the key lessons learned was the importance of data quality in the analytics lifecycle. A significant portion of the project effort was spent understanding the datasets, handling missing values, and standardizing fields before meaningful analysis could begin. The project also reinforced the value of combining technical analysis with business interpretation, as data insights become more impactful when translated into actionable recommendations. Overall, the assignment provided valuable experience in building data pipelines, performing exploratory analysis, and applying analytical thinking to solve real-world business problems.

Appendix A – AI Usage Disclosure

AI Tools Used

- ChatGPT (OpenAI)
- GitHub Copilot
- Claude Ai

Sections Assisted by AI

AI tools were used to assist with:

- Report structure and formatting
- Code review and debugging
- Documentation drafting
- SQL query refinement
- Interpretation of analytical results

Example Prompts Used

- "Design a star schema suitable for analyzing Airbnb booking and listing data."
- "Review the data cleaning logic and suggest improvements for accuracy and consistency."
- "Interpret the exploratory data analysis findings from a business perspective."
- "Recommend appropriate evaluation metrics for the machine learning models used in this project."
- "Draft professional report sections covering the data engineering and analytics workflow."

Validation Process

All AI-generated outputs were manually reviewed, tested, and validated before inclusion in the final submission. Code suggestions were executed and verified against the dataset, while report content was checked for accuracy and alignment with project findings. Final decisions regarding methodology, analysis, interpretations, and recommendations were made by the author.